

Hardware-Aware Quantum Materials Simulation on IBM Quantum

Project proposal · UOttawa Qiskit Fall Fest 26 · 3–4 day build

Goal

Improve the documented limitations of our previous Quantinuum project (1D transverse-field Ising model and 2D Fermi–Hubbard; exact diagonalization, Trotter, VQE, ZNE, Iceberg error detection). We build a hardware-aware simulation and error-mitigation workflow on IBM Quantum hardware and test whether it is more accurate and efficient than the previous approach.

Why Iceberg on heavy-hex is not our main path

- Every logical gate uses a hub qubit ($\tilde{X}_i = X_t X_i$, $\tilde{Z}_i = Z_b Z_i$) and both syndrome ancillas must reach all data qubits: a star graph. Heavy-hex qubits have at most 3 neighbours, so each Trotter step needs $O(k)$ -depth SWAP networks.
- The code has distance 2, and a failing SWAP is a correlated two-qubit error, so SWAPs create errors the code cannot detect.
- Iceberg's advantage relies on all-to-all connectivity and native MS gates (trapped ions); even there the previous report saw 47–95% discard rates.

Decision: use hardware-native detection and mitigation on IBM hardware; run Iceberg only as a simulated cost-benefit study (all-to-all vs heavy-hex).

Approach: limitation → fix → evidence

Previous limitation	Planned fix (IBM / Qiskit)	Evidence of improvement
Iceberg discard rate 47–95%	Ancilla-free symmetry post-selection (particle number for Hubbard, parity for TFIM)	Discard rate and error vs report
ZNE overcorrects after $t \approx 1.5$	Runtime Estimator: Pauli twirling, dynamical decoupling, TREX readout, ZNE with exponential extrapolation	Error vs time vs linear 3-point ZNE
Trotter error at $h/J = 2$ (8.46% at $\Delta t = 0.05$)	Multi-product formulas (qiskit-addon-mpf)	<5% at $h/J = 2$ with no deeper circuits
Depth and gate-count limits	Native RZZ gates, edge-coloured layers, 12-qubit ring mapped natively onto heavy-hex	2-qubit gate count and depth tables
Iceberg + ZNE cost-benefit left open	Aer with Heron-like noise; Iceberg transpiled to all-to-all vs heavy-hex	SWAP overhead and break-even depth plot
VQE does not converge	Deprioritized; adiabatic preparation + MPF (SQD as stretch goal)	n/a

3–4 day plan

1. **Day 1:** Qiskit port of exact diagonalization and Trotter circuits, validated against the previous report's Tables 1–2; noise model, mitigation pipeline, MPF code.
2. **Day 2:** Simulation studies (ZNE extrapolators, MPF vs 2nd-order Trotter, Iceberg topology study); freeze hardware circuits.
3. **Day 3:** Batched hardware runs for the TFIM; 2×2 Hubbard with post-selection if budget allows.
4. **Day 4:** Plots, comparison table vs the previous report, write-up.

Risks and early checks

- **QPU time:** Open Plan gives ~10 min/month; confirm the hackathon allocation and batch jobs.
- **Fractional gates:** verify twirling/PEA support with native RZZ; keep a CZ/ECR fallback.
- **2D Hubbard:** Jordan–Wigner strings cost many SWAPs on heavy-hex; limit to 2×2 or use a 1D chain.

Deliverables

Reproducible Qiskit repository, hardware and simulation results, and a table comparing accuracy, discard rate and circuit cost against the previous Quantinuum results: lower Trotter error without deeper circuits, better mitigation than linear ZNE, and a quantified answer on when Iceberg pays off on heavy-hex.